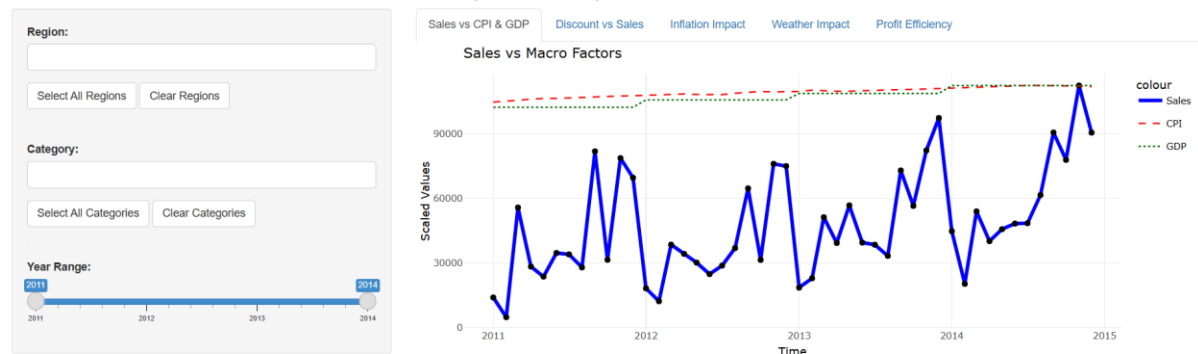


Superstore-Retail-Analysis

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R SHINY DASHBOARD

US Retail Performance vs External Factors (2011–2014)



Problem Statement

Retail performance is influenced by a combination of internal business decisions and external environmental conditions. While organisations can control factors such as pricing, discount strategies, and logistics, broader macroeconomic and environmental forces also play a significant role in shaping consumer behaviour and demand.

This study uses the Superstore dataset (2011–2014) to examine patterns in sales, profit, and discount effectiveness across time and regions within the United States. Although the dataset provides detailed transactional information, it does not explain the underlying reasons for observed fluctuations in performance.

To address this limitation, the study incorporates external datasets on inflation (Consumer Price Index [CPI]), economic development (GDP per capita), and weather conditions. These variables are used to provide additional context for understanding changes in retail outcomes.

Accordingly, this study investigates the extent to which macroeconomic and environmental factors help explain variations in retail performance beyond internal operational factors alone.

Dataset Selection and Justification

1. Consumer Price Index (CPI)

Source: Federal Reserve Economic Data (FRED), U.S. Bureau of Labor Statistics

<https://fred.stlouisfed.org/series/CPIAUCSL>

Brief Description

The Consumer Price Index (CPI) measures changes in the price level of a fixed basket of goods and services over time, reflecting inflation and consumer purchasing power.

Justification

CPI was selected because inflation directly affects consumer spending behaviour and price sensitivity in retail markets.

Complement to Superstore Dataset

The Superstore dataset contains transactional data (sales, profit, discounts) but lacks economic context. CPI provides this by showing how inflation may influence retail performance over time (2011–2014).

2. GDP per Capita (World Bank)

Source: World Bank Open Data

<https://data.worldbank.org/indicator/NY.GDP.PCAP.CD>

Brief Description

GDP per capita measures average economic output per person and reflects a country's level of economic development and purchasing power.

Justification

It was chosen because higher national income levels are generally associated with stronger consumer spending and retail demand.

Complement to Superstore Dataset

While Superstore data shows regional sales and profit, it does not explain economic differences between locations. GDP per capita adds this macroeconomic context.

3. Weather Data (Temperature & Precipitation)

Source: Open-Meteo Historical Weather API

<https://open-meteo.com/>

Brief Description

This dataset includes historical temperature and precipitation data that reflect environmental conditions over time.

Justification

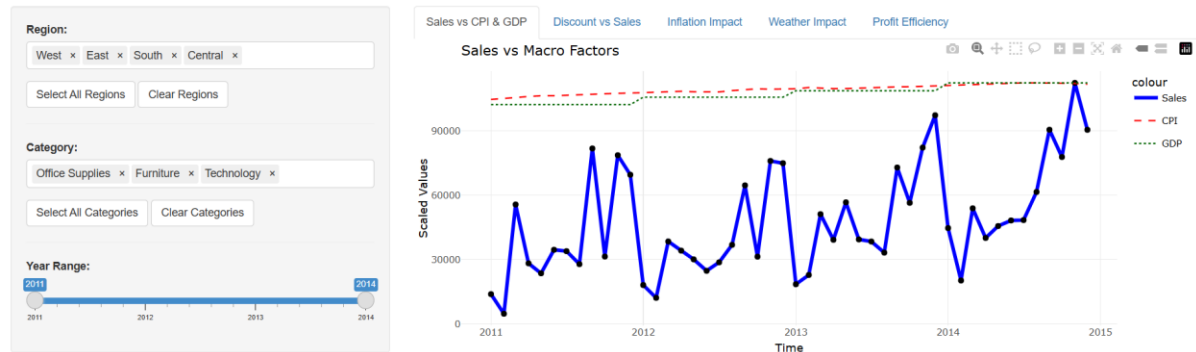
Weather influences short-term consumer behaviour, affecting shopping frequency and demand patterns.

Complement to Superstore Dataset

Superstore data does not include environmental factors. Weather data helps explain seasonal and climate-related variations in sales.

P1: Sales vs CPI & GDP (Macroeconomic Trends)

US Retail Performance vs External Factors (2011–2014)



What the visualization shows

This line chart compares monthly Superstore sales in the United States from 2011 to 2014 against two macroeconomic indicators: the Consumer Price Index (CPI) and GDP per capita. CPI represents inflationary pressure within the economy, while GDP per capita reflects overall economic development. Both indicators are scaled to enable meaningful comparison with sales trends over time.

Interpretation

The results indicate that Superstore sales fluctuate considerably over the observed period, with clear seasonal peaks and troughs. In contrast, CPI demonstrates a gradual and consistent upward trend, reflecting steady inflation, while GDP per capita shows slow but stable economic growth.

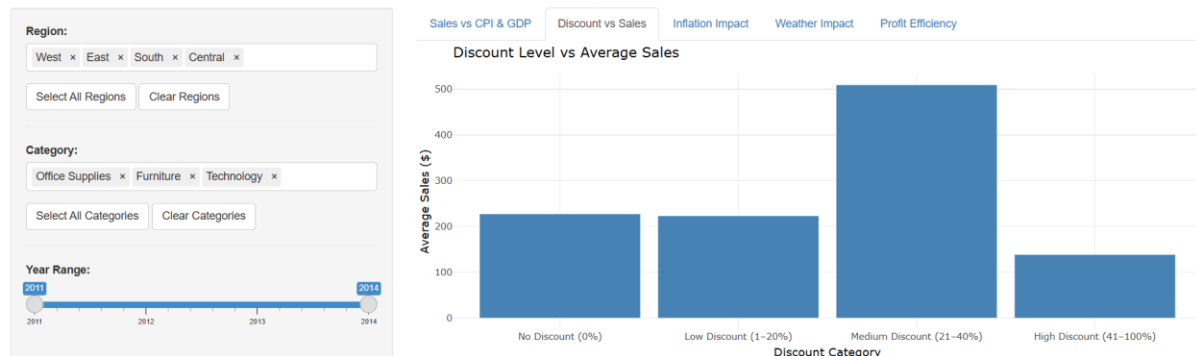
However, sales movements do not consistently align with either CPI or GDP trends. This suggests that while macroeconomic conditions may influence consumer purchasing power in a broad sense, they do not directly explain short-term variations in retail sales. Instead, sales appear to be more sensitive to seasonal effects and internal business dynamics.

Contribution to the problem statement

This visualization supports the research objective by demonstrating that macroeconomic indicators provide important contextual background but are insufficient on their own to explain fluctuations in retail performance. It highlights that external economic conditions contribute to overall trends but do not fully account for short-term sales variability in the Superstore dataset.

P2: Average Sales by Discount Level

US Retail Performance vs External Factors (2011–2014)



What the visualization shows

This bar chart compares average sales across four discount categories: no discount, low discount, medium discount, and high discount. It illustrates how pricing strategies influence consumer purchasing behaviour.

Interpretation

The results indicate a non-linear relationship between discounting and sales performance. Moderate discount levels are associated with the highest average sales, whereas both no discount and high discount strategies are linked to lower performance outcomes.

This pattern suggests that excessive discounting does not necessarily stimulate higher demand and may reduce perceived product value. Conversely, moderate discounts appear to strike an effective balance between attracting customers and maintaining perceived product worth.

Contribution to the problem statement

This finding highlights the importance of internal business decisions in shaping retail performance. In particular, it demonstrates that pricing strategy has a strong influence on sales outcomes, often exceeding the explanatory power of external macroeconomic conditions in the dataset.

P3: Sales Distribution Across Inflation Levels (CPI Boxplot)

US Retail Performance vs External Factors (2011–2014)



What the visualization shows

This boxplot illustrates the distribution of sales across three inflation environments: low, medium, and high CPI periods. Unlike time-series plots, it focuses on variability and distributional differences in sales under different inflation conditions.

Interpretation

The median sales values remain relatively consistent across all inflation categories, suggesting that inflation does not substantially alter average sales levels in the short term. However, higher inflation periods exhibit a wider spread of sales values, indicating greater variability in retail performance.

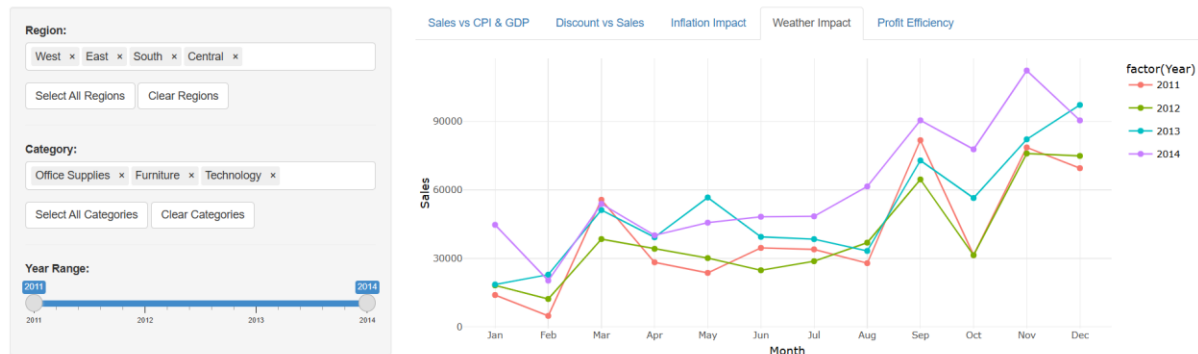
This pattern suggests that while inflation may not significantly shift demand levels, it may contribute to increased uncertainty in consumer purchasing behaviour.

Contribution to the problem statement

This visualization refines the understanding of macroeconomic influence by showing that inflation affects the stability of retail performance rather than its overall direction. It suggests that external economic conditions may introduce variability without directly driving sales increases or decreases.

P4: Seasonal Sales Patterns and Weather Impact

US Retail Performance vs External Factors (2011–2014)



What the visualization shows

This line chart presents monthly sales trends from 2011 to 2014, disaggregated by year, alongside average temperature patterns. It is designed to explore potential relationships between environmental conditions and seasonal retail performance.

Interpretation

The results reveal clear and recurring seasonal patterns in sales, with consistent peaks and declines observed across years. These patterns indicate that consumer demand is strongly influenced by seasonality.

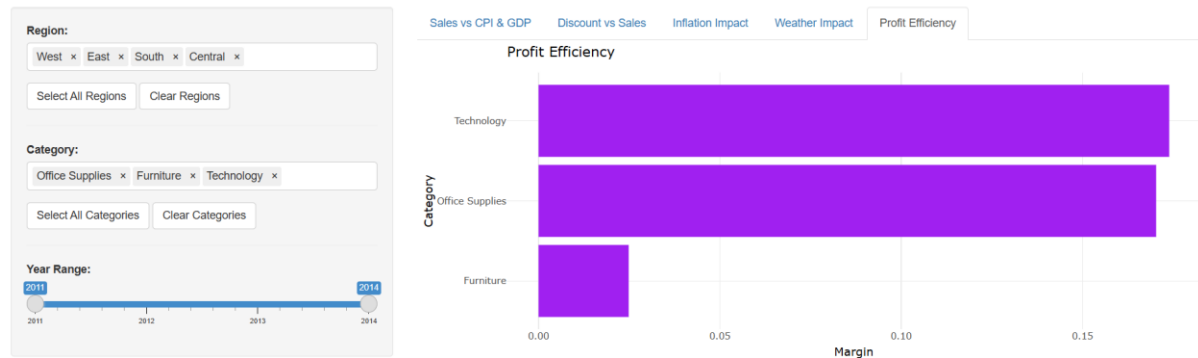
Temperature varies across months in a predictable seasonal cycle; however, its relationship with sales is not strictly proportional. Instead, weather appears to reinforce existing seasonal demand patterns rather than independently driving sales changes.

Contribution to the problem statement

This visualization demonstrates that environmental conditions contribute to predictable seasonal fluctuations in retail performance. It supports the broader argument that external factors influence the timing and rhythm of sales, rather than directly determining overall sales magnitude.

P5: Profit Efficiency by Product Category

US Retail Performance vs External Factors (2011–2014)



What the visualization shows

This bar chart compares profit margins across product categories by measuring profit relative to sales. It evaluates how efficiently each category converts revenue into profit.

Interpretation

The results indicate substantial variation in profitability across product categories. Some categories generate high sales volumes but relatively low profit margins, while others achieve stronger profitability with lower sales.

These differences suggest that revenue alone is not a sufficient indicator of business success. Instead, structural factors such as cost composition, pricing strategy, and product characteristics play a critical role in determining profitability.

Contribution to the problem statement

This visualization addresses the internal operational dimension of the research problem. It demonstrates that retail performance is not solely driven by sales volume but also by profit efficiency, which is primarily shaped by internal business decisions rather than external macroeconomic conditions.

Data Story: Retail Performance Analysis (2011–2014)

What is happening overall?

The analysis of the Superstore dataset indicates that retail performance is shaped by both internal business decisions and external macroeconomic conditions. Across 2011–2014, sales and profit exhibit noticeable fluctuations, including strong seasonal patterns and variation across product categories.

However, when external variables such as inflation (Consumer Price Index [CPI]), economic growth (GDP per capita), and weather conditions are incorporated, it becomes evident that these factors primarily provide contextual background rather than direct explanations for short-term performance changes. This aligns with macroeconomic theory, which suggests that aggregate indicators influence long-run consumption trends but have limited power in explaining firm-level volatility (Mankiw, 2021; Blinder, 1997).

Instead, internal operational decisions appear to play a more immediate and direct role in shaping retail outcomes.

How do macroeconomic conditions relate to sales?

The comparison between sales, CPI, and GDP per capita shows that retail performance does not closely follow macroeconomic trends. While GDP per capita increases gradually over time and CPI reflects steady inflationary pressure, sales demonstrate higher volatility and frequent short-term fluctuations.

Although economic growth is generally associated with increased consumer purchasing power, it does not appear to directly explain monthly changes in sales. This is consistent with economic literature suggesting that macroeconomic indicators primarily affect aggregate demand rather than short-term retail behaviour (Dornbusch et al., 2018).

In this context, CPI and GDP function more as background indicators of economic conditions rather than direct drivers of sales performance.

How does pricing strategy influence performance?

The analysis of discount levels reveals a clear non-linear relationship between pricing strategy and sales outcomes. Moderate discount levels are associated with the highest average sales, while both no discount and high discount strategies result in weaker performance.

This pattern supports marketing theory, which suggests that pricing affects both demand and perceived product value (Kotler & Keller, 2016). Excessive discounting may reduce perceived quality or profitability, while insufficient discounting may fail to attract price-sensitive consumers. Similar findings in promotional effectiveness research indicate that moderate promotions often outperform extreme pricing strategies (Blattberg et al., 1995).

Overall, pricing strategy emerges as a strong internal driver of consumer behaviour and appears more directly influential than macroeconomic conditions in shaping sales outcomes.

How does inflation affect retail behaviour?

The analysis of sales across different inflation levels shows that median sales remain relatively stable across low, medium, and high inflation periods. However, higher inflation environments exhibit slightly greater variability in sales outcomes.

This suggests that inflation does not significantly alter overall demand levels within the period studied but may increase uncertainty in consumer purchasing behaviour. Economic theory supports this interpretation, noting that inflation primarily affects real purchasing power and consumer expectations rather than immediately reducing consumption in stable economic environments (Dornbusch et al., 2018).

Thus, inflation appears to influence the consistency of sales rather than their overall direction.

What role does seasonality and weather play?

The seasonal analysis reveals strong and consistent patterns in sales across all years, with predictable peaks and troughs occurring at similar times annually. This indicates that consumer demand is highly influenced by seasonal cycles.

While weather conditions such as temperature follow expected seasonal variations, their direct relationship with sales is relatively weak. Instead, weather appears to reinforce existing seasonal demand patterns rather than independently driving changes in consumer behaviour.

This aligns with consumer behaviour research, which shows that purchasing activity is often structured around seasonal and calendar-based cycles rather than short-term environmental changes alone (Agrawal & Schor, 2018).

Therefore, environmental conditions primarily influence the timing of demand rather than its magnitude.

Why do some product categories perform better than others?

The analysis of profit efficiency across product categories reveals substantial variation in profitability. Some categories generate high sales volumes but relatively low profit margins, while others achieve stronger profitability with lower overall sales.

This indicates that revenue alone does not fully capture business performance. Differences in cost structures, pricing strategies, and product characteristics significantly influence profitability outcomes.

From a strategic perspective, this aligns with Porter's (1985) theory of competitive advantage, which argues that firm performance is strongly shaped by internal structure and strategic positioning rather than external market conditions alone.

Why is this happening overall?

When all findings are considered together, a consistent pattern emerges. External macroeconomic and environmental factors provide useful contextual information but do not fully explain short-term fluctuations in retail performance.

Specifically:

- Macroeconomic indicators (CPI and GDP) provide long-term economic context but have limited short-term predictive power (Mankiw, 2021; Blinder, 1997)
- Inflation affects variability in outcomes rather than overall demand levels (Dornbusch et al., 2018)
- Weather and seasonality shape timing patterns rather than sales magnitude (Agrawal & Schor, 2018)
- Internal business decisions, particularly pricing strategy and product structure, have the strongest direct influence on performance (Kotler & Keller, 2016; Porter, 1985)

Together, these findings suggest that retail performance is best understood as a system where internal operational decisions dominate short-term outcomes, while external conditions provide secondary explanatory context.

How should the business respond?

The findings suggest several practical implications grounded in both data and theory:

- Pricing strategies should focus on optimal discount levels rather than excessive promotions, consistent with promotional effectiveness theory (Blattberg et al., 1995)
- Profitability should be evaluated at the product category level, reflecting strategic differences in cost and value creation (Porter, 1985)
- Macroeconomic indicators should be used as contextual planning tools rather than direct forecasting variables (Mankiw, 2021)
- Seasonal patterns should guide demand forecasting, inventory planning, and marketing cycles (Agrawal & Schor, 2018)
- Greater emphasis should be placed on internal decision-making, as it has the strongest influence on performance outcomes

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Artificial Intelligence (AI) Acknowledgement

Artificial intelligence tools (ChatGPT by OpenAI) were used to assist in identifying and locating relevant external datasets used in this study, including macroeconomic (CPI and GDP per capita) and environmental (weather) data sources. All datasets were subsequently verified and obtained from official and publicly available data repositories.